

Shivam Rajput

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EDUCATION

AISSMS IOIT

B.Tech in Computer Engineering

Pune, Maharashtra

Nov. 2022 – May 2026

PROJECTS

Quack - Search Engine | Python, Node.js (Express), MongoDB, Redis, NLTK, REST APIs, TF-IDF

[GitHub](#)

- Built a multi-threaded search engine pipeline with a **64-thread web crawler** that collects and normalizes up to 20k pages per crawl cycle.
- Designed an inverted index with **TF-IDF** scoring and PageRank ranking, computing link graph scores using sparse matrix power iteration.
- Implemented **Redis**-based staging for posting lists to avoid MongoDB array rewrites, improving indexing throughput.
- Developed a REST search API with query parsing, stemming, and caching, serving ranked results with snippet extraction, achieving an average query response time of **0.02s**.

OmniView - AI Satellite Intelligence Platform | Python, Flask, Electron, TensorFlow, PyTorch

[GitHub](#)

- Built a geospatial desktop platform (**Electron + Flask**) for real-time satellite imagery analysis supporting road extraction and multi-temporal change detection.
- Developed a **road extraction pipeline** using a fine-tuned **ResNet model** (TensorFlow), processing large Sentinel-2 TIFF tiles via a 2×4 crop-grid inference strategy, generating probability and binary road masks with an integrated **Hugging Face** Space API fallback.
- Implemented a **U-Net** (PyTorch) for built-up area **change detection** between multi-temporal satellite images, using 6-channel stacked input (two 3-band scenes) with encoder-decoder skip connections.

Brain Tumor Detection | React, Node.js, MongoDB, Hugging Face, Gemini API, Clerk

[GitHub](#)

- Built a **full-stack** web app that analyzes brain MRI scans to detect tumors and classify their type, using a custom CNN model (deployed on Hugging Face) with a MERN-based frontend and backend.
- Implemented **real-time predictions** with automated tumor summaries using Gemini API, generating automated tumor summaries alongside predictions to assist interpretation.
- Integrated Clerk authentication and MongoDB **persistence** to ensure secure, personalized usage across sessions.

EXPERIENCE

Web Development Intern

Dec 2024 – Feb 2025

Edunet Foundation

Remote

- Developed a full-stack music streaming platform using the **MERN stack**, implementing core features including real-time audio playback, playlist management, and personalized user libraries.
- Designed and integrated **JWT-based authentication** with secure session handling, enabling role-based access and persistent user state across sessions.
- Architected a **RESTful API** with MongoDB for media metadata storage and deployed the application on **Render** with environment-based configuration for production readiness.

TECHNICAL SKILLS

Languages: C++, Python, JavaScript

Frontend: React, HTML, CSS

Backend & Libraries: Node.js, Express.js, JWT, Socket.io

ML / AI: TensorFlow, PyTorch, NLTK, Hugging Face, Gemini API

Databases: MongoDB, MySQL, Redis

DevOps & Tools: Docker, Git, GitHub, Postman, Linux

Cloud & Deployment: Azure, Vercel, Render, Hugging Face Spaces

CERTIFICATIONS

Microsoft Certified: Azure Fundamentals (AZ-900) – Microsoft, 2026
Oracle Cloud Infrastructure Foundations Associate – Oracle University, 2025
MERN Stack Development – Edunet Foundation, 2025

ACHIEVEMENTS & OPEN SOURCE

Top 10 – National Level PBLM Competition (2023): Built **Aashray**, a platform connecting homeless individuals with nearby NGOs for emergency shelter and aid; competed against **200+ teams** nationwide.
AI Summit (2025): Showcased **OmniView**, an AI-powered satellite intelligence platform, demonstrating real-time road extraction and change detection to an audience of Industry professionals.
Hacktoberfest 2024 Contributor: Submitted accepted pull requests to open-source repositories, demonstrating collaborative coding and version-control best practices.